SSHVisual: Visual Tool to Analyze SSH Data

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Abstract

Attackers attempt to log in to computer systems or servers all over the world in order to use them for nefarious purposes. We have collected the login attempts by these attackers on our servers over SSH. Previous works in network visualization have 1. analyzed the attackers' IP and AS dimension of the SSH logins, 2. used automated detection systems to find anomalous network events given baseline events and, 3. employed collaborative methods to help analysts share findings about network events with each other. With our visualization SSHVisual, we 1. analyze the IP and AS and give more attention to the username dimension of SSH attacker logins to draw a holistic picture of SSH attackers in service of SSH campaign detection, 2. utilize detection by analysts of anomalous events in absence of baseline events, 3. employ collaboration like previous works, 4. use a novel similarity metric to find out attacker persistence over time and 5. correlate SSH attacks with real-world events. We build all these features in SSHVisual by studying tasks and requirements of network administrators. Using case studies with network admins, we demonstrate that our approach helps them find important events from data.

I Introduction

Computer systems worldwide employ SSH to allow their users to securely connect to them in order to execute commands and do remote administration. A lot of users do a username, password combination (password based authentication) in order to gain entry to the SSH server with key based authentication being less common. Attackers, who are adversarial in nature, can use brute force attack mechanisms to get into SSH servers. Sometimes, they guess what the username, password combinations are and other times, they take advantage of poorly secured servers. Then, these attackers use these servers to launch botnet campaigns to carry out further attacks or steal resources from the servers, like cryptomining.

Knowing the prevalence of SSH attacks in the IPv4 space, we decided to collect SSH data with our servers with the purpose of understanding attacker behavior visually and holistically. To that end, we built a visual tool called SSHvisual in order to address the gaps that existing network visualizations have and use best practices that network visualizations already employ. Surveys of network visualizations [1]–[5] establish both the maturity of the network visualization field and the gaps we target.

Three visual contributions address the gaps we have found. First, actor-level analysis: we treat the attacker as a first-class actor — including the username identity, not only the source IP or autonomous system (AS) — and we identify coordinated campaigns, since our data affords no stable baseline for automated detection. Second, we apply a novel similarity metric to these actors, particularly in the username dimension, as they evolve over time. Third, we corroborate actor behavior in the visualization with real world events, with a particular emphasis on the username dimension given the limitations of IP corroboration. Additionally, following established practices, we allow for analyst driven collaboration and labeling of irregularities found in the data.

II Related Work

A long line of tools [6, 7, 8, 10, 27, 12, 21, 13] render network traffic and security events navigable for operators and analysts with the majority being network flow and traffic oriented. Additionally, a smaller alert- and log-oriented group [9, 23, 26, 15, 16, 11, 18, 22, 19, 20] organizes IDS output and heterogeneous logs of which [15] contributes a visual analysis (of the CICIDS2017 dataset) that isolates attack clusters — including SSH brute-force (SSH-Patator) — through clustering and dimensionality reduction, namely a multi-resolution wavelet-feature analysis.

A Attack Campaigns (Emphasis on the Username Dimension) with Time

Honeypot literature does not promote the username identity as an actor in its own right (for attack campaign detection purposes): it remains an attribute to count, never a unit to track — the unit that, alongside the IP and AS, anchors our analysis. Some papers [30, 31, 32, 34, 35, 36, 38] collect and chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted — the SCADA SSH honeypot [34], kippo-graph [35], and the container deployment [36] — usernames and passwords appear only as static top-N tallies (kippo-graph's top-10 username and password bars are the archetype), never placed on a time axis even where the same tool charts attack volume over time.The WAN honeypot framework [37] is the closest prior system on the credential axis: it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist — a manual, one-off prototype of credential-based coordination analysis, performed on the IP unit and without a temporal dimension.

B Attacker Similarity (Emphasis on the Username Dimension) with Time

Computed similarity between entities is well established in literature [14, 24, 28, 29]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on and how it moves through time. On the entity: these systems cluster internal hosts [14], behavioral profiles [28], IP/port members [29], or internal enterprise users grouped by shared resources — not external attacker identities [24]. None treats the username as an attacker actor. On time: [29, 28] reach into the temporal dimension, but both do not compute the similarity of the attacker population across consecutive windows. We apply similarity based metrics — for example, Jaccard overlap of attackers between successive windows — particularly to the username actor unit, capturing how the attacker population persists or churns over time.

C Limits of automated detection

A central premise of our approach is that SSH traffic does not admit the "normal baseline" that automated anomaly and campaign detection presuppose. The broader literature documents why such detection is fragile: Sommer and Paxson [M1] note that, despite extensive research, machine-learning anomaly detection is rarely deployed for intrusion detection, given the enormous variability of "normal," the high cost of errors, and the difficulty of sound evaluation; Axelsson [M2] shows that automated anomaly detection is throttled by the base-rate fallacy: where intrusions are rare relative to benign traffic, the false-positive rate becomes the limiting factor. Honeypot traffic departs from that setting entirely — it offers no stable 'normal' against which to define an anomaly, with Gates and Taylor [M3] supporting our approach by challenging the paradigm directly, questioning that anomalies and attacks are rare, that a stable normal exists and sometimes normal traffic may be malicious and malicious traffic may be normal, finally observing that detection ultimately devolves to an analyst manually investigating clusters of activity.

This fragility is visible from some of the papers referenced so far. The automated detectors there are built on a per-actor baseline: Situ [14] learns a probabilistic model of each IP's behavior and flags low-probability events, and CLIQUE [25] scores each actor's deviation from a calculated baseline of its own expected activity. Both presume the stable per-actor "normal" that honeypot traffic — near-uniformly hostile and constantly churning — cannot supply, so they do not transfer to this setting by construction. And even where such a baseline does hold, Situ itself concedes that automation is not enough: having reviewed the limits of signature-, supervised-, and reputation-based detection, it argues that operators "do not want another automated tool with algorithms they do not trust," and builds a human-in-the-loop visual analytics system instead. We reach the same conclusion from a starker starting point: lacking even the per-actor baseline these systems assume, we treat anomaly and campaign detection as analyst-driven from the outset, providing visual support for it rather than another automated detector.

D Actor identity and real-world event correlation

The attacks we observe do not arrive at a steady rate; they rise, fall, and change in character as events unfold in the wider world, so understanding them means lining them up against what is happening outside the network. The link is well documented: Bilge and Dumitras [M7] found that the number of attacks against a given vulnerability can jump by up to five orders of magnitude — a hundred-thousand-fold increase — once it is publicly disclosed, and Sabottke et al. [M8] showed that future exploitation can be predicted from signals in disclosure channels alone. Allodi and Massacci [M9] add that a vulnerability's official severity score is a poor guide to whether it is actually attacked, so the dependable way to learn what attackers are reacting to is to watch real activity and line it up against real events, not to read it off a score. We want to do this at the level of a campaign — a group of attackers acting together — rather than one attacker at a time, and that raises a practical problem: a campaign rarely shows up as a single source address. IP addresses do not stay tied to one machine — one host is assigned many addresses over time [M4], hosts cycle through new addresses [M5], and counting by address can overstate the number of real machines by orders of magnitude [M6] — so it is likely that a view organized by address scatters one campaign across dozens of entries that look unrelated. The username pulls those pieces back together: the same login names likely recur across the changing addresses, so grouping by username re-assembles the campaign that an address-by-address view splits apart. A username is not a perfect identity either — names like "root" are tried by almost everyone, and one campaign runs through a whole list of names — so we treat it as one analysis dimension alongside the address and the network it belongs to (its AS), resolving the overlaps with the similarity and grouping methods above.

E Analyst labeling and collaboration (Emphasis on the Username Dimension)

Because detection on this data is analyst-driven, the findings an operator produces by hand must be captured and shared rather than discarded. In two, lines of work [17, 21], interactive labeling has been used to fold the analyst's judgments back into the system. Collaborative analysis has been supported through shared artifacts with [12] and [27] being notable examples. These systems establish that labeling and collaboration are valuable, but each works on a different kind of data and a different unit: none lets an analyst group username-level campaign actors observed in SSH login traffic into a named, persistent artifact that other operators can see and build on. Our labeling-and-sharing mechanism is the step that turns a manually identified set of usernames into a durable, shared campaign.

Table 1: Example placeholder table

Requirement	Task 1	Task 2	Task 3	Task 4
Row A	+	-	+	+
Row B	tech	plan	-	+
Row C	+	+	+	-

III Typical Tasks and Visualization Requirements

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A Data and Domain Characterization

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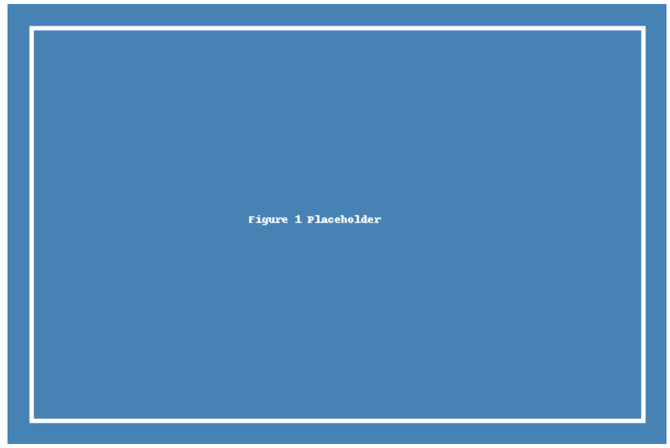

Figure 1: Show campaign detection using username+IP+AS

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IV Approach

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A First Component

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B Second Component

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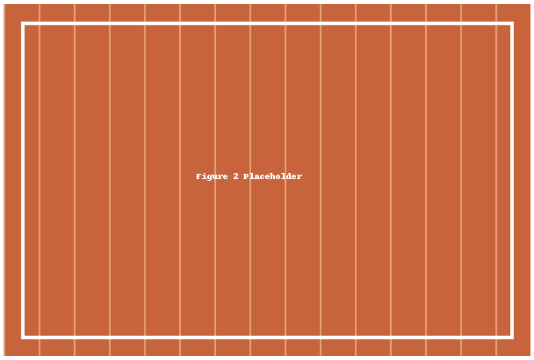

Figure 2: Show collaboration + labeling

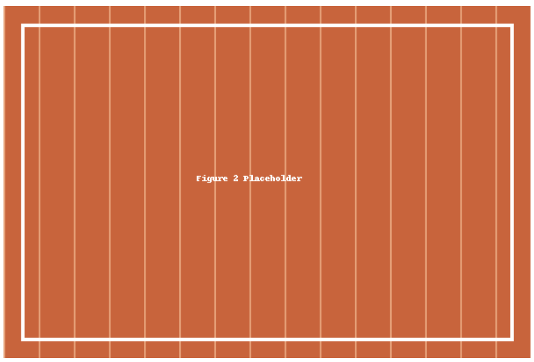

Figure 3: Show similarity metric at play

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C Third Component

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V Implementation

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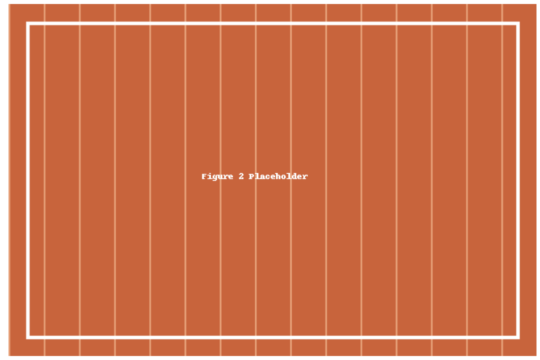

Figure 4: Show correlation of attacks with real events

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A System Architecture

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B Data Pipeline

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VI Evaluation and Discussion

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A Case Studies

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B Discussion

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VII Summary

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A Limitations

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B Future Work

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